# PRACTICE SHEET

**1.** Consider the following statements:
I. If $\theta = 1200°$, then $(\sec\theta + \tan\theta)^{-1}$ is positive
II. If $\theta = 1200°$, then $(\operatorname{cosec}\theta - \cot\theta)$ is negative.
Which of the statements given above is/are correct?
(a) Only I (b) Only II
(c) Both I nor II (d) Neither I nor II

**2.** Match List I with List II and select the correct answer using the code given below the lists.

| | List I | | List II |
|---|---|---|---|
| A. | tan15° | (i) | $-2-\sqrt{3}$ |
| B. | tab75° | (ii) | $2+\sqrt{3}$ |
| C. | tan105° | (ii) | $-2+\sqrt{3}$ |
| | | (iv) | $2-\sqrt{3}$ |

(a) A-(iv), B-(i), C-(ii) (b) A-(iv), B-(ii), C-(i)
(c) A-(iii), B-(ii), C-(i) (d) A-(ii), B-(i), C-(iv)

**3.** If $A+B+C=\frac{\pi}{2}$, then what is the value of tan A tan B + tan B. tan C + tan C tan A?
(a) 0
(b) 1
(c) −1
(d) tan A tan B tan C

**4.** What is the value of

**8.** If $\sin x + \operatorname{cosec} x = 2$, then what is the value of $\sin^4 x + \operatorname{cosec}^4 x$?
(a) 2 (b) 4
(c) 8 (d) 16

**9.** What is the value of $\tan(-1575°)$?
(a) 1 (b) ½
(c) 0 (d) −1

**10.** For which acute angle θ, $\operatorname{cosec}^2\theta = 3\sqrt{3}\cot\theta - 5$?
(a) $\frac{5\pi}{12}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$

**11.** If $\tan^2\theta = 2\tan^2\phi + 1$, then which one of the following is correct?
(a) $\cos(2\theta) = \cos(2\phi) - 1$
(b) $\cos(2\theta) = \cos(2\phi) + 1$
(c) $\cos(2\theta) = [\cos(2\phi) - 1]/2$
(d) $\cos(2\theta) = [\cos(2\phi) + 1]/2$

**12.** If $\cos\theta < \sin\theta$ and θ lies in the first quadrant, then which one of the following is correct?
(a) $0 < \theta < \frac{\pi}{4}$ (b) $\frac{\pi}{4} < 0 < \frac{\pi}{2}$